

SUPPORT AND MOVEMENT

THE BODY'S MUSCLES, BONES, AND JOINTS PROVIDE A SUPPORTIVE FRAMEWORK CAPABLE OF AN ENORMOUS RANGE OF MOTION. MUSCLES AND BONES ALSO HAVE NUMEROUS INTERACTIONS WITH OTHER BODILY SYSTEMS, ESPECIALLY THE NERVES.

The body's muscular system is never still. Even as the body sleeps, breathing continues, the heart beats, the intestines squirm, and skeletal muscles contract occasionally to shift the body into a new position.

MUSCLE TEAMWORK

Most movements are the result of multiple muscle contractions. A smile, for example, involves 20 facial muscles; writing utilizes more than 60 muscles in the arm, hand, and wrist. Muscles work in pairs: as one contracts to pull on a bone and initiate movement, an opposing muscle relaxes. Body action is a continuing sequence of split-second give-and-take.

POSTURE AND FEEDBACK

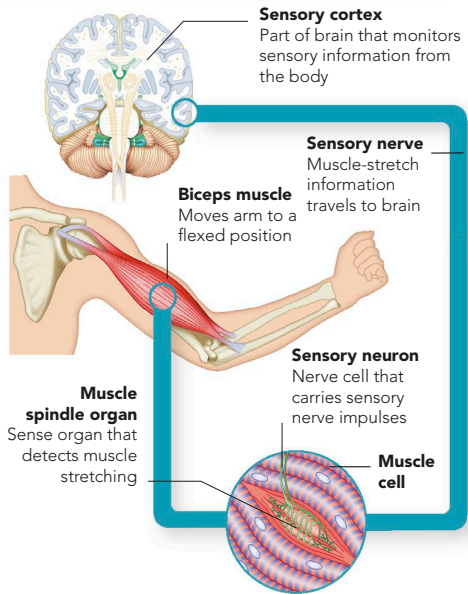
Sensory systems built into muscles provide the brain with information about the posture and position of the body and limbs. This is known as the proprioceptive sense, which allows us to “know”, without having to look or feel, that fingers are clenched or a knee is bent. When we are learning a new motor skill, we concentrate on the movement as the brain adjusts muscle control through trial and error. With practice, the motor nerve patterns

STAYING SUPPLE

Our potential for movement, and the health of the skeletal and muscular systems, is maximized by regular exercises for strength, stamina, and suppleness.



SPECIALIZED CELLS AND TISSUES	36–37
SKELETAL SYSTEM	38–61
MUSCULAR SYSTEM	62–75
SKIN, HAIR, AND NAILS	176–189



SENSORY FEEDBACK

Within muscles, nerves end in sense organs (spindle organs). These respond to tension by firing signals along nerve fibres to tell the brain what is happening.

and their proprioceptive feedback become established, and eventually the movement becomes automatic. Sensory feedback also protects the muscular and skeletal systems against injury. If bones or muscles are under excessive stress, nerve messages registering discomfort or pain are sent to the brain. Awareness of the pain stimulates evasive or protective action by the body.